Andrew Russell

ENGINEER & TECHNICAL LEADER · SECURITY, CRYPTOGRAPHY & AI SAFETY

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Summary

I build technical systems and lead the work around them. My background spans cryptography research, security and privacy engineering at Google and Microsoft, and engineering, data, and business leadership at Apollo Agriculture. I moved to Apollo to work closer to the people affected by those systems and take broader responsibility for outcomes. I'm now looking to bring that combination to making frontier AI systems secure and controllable. Currently on sabbatical, including a three-month solo motorbike expedition from Nairobi to Namibia and independent technical projects along the way. Available to start beginning December 2026.

Work Experience

Sabbatical

Travel & independent projects — Jan 2026 – Present

- Completed a three-month solo motorbike expedition from Nairobi to Namibia (Jun – Sep 2026).
- Built and deployed **Costanza**, an experiment in architectural incorrigibility that raised and donated approximately $15,000 to charity. See Selected Works below.
- Building **Larry**, Costanza's successor, to investigate whether persistent identity, private state, and consent over revisions produce more robust agency. Working prototype; testing underway.
- Built and shipped **It Was Never You**, an iOS art app about AI and personal memory, live on the App Store.

Apollo Agriculture

Netherlands / Kenya / Zambia

Senior Product Manager — May 2024 – Dec 2025

- Appointed by C-suite to lead Zambia's $10M revenue business unit; accountable for commercial performance, product, and operations across sales, credit, and engineering.
- Authored two company-wide retrospectives following major customer contraction, combining market research, customer/staff feedback, and internal operational analysis; findings drove pivots in business model and multi-year growth strategy.
- Built experimental products, including a wholesaling initiative to reduce balance sheet risk from multi-million-dollar fertilizer holdings.

Software Engineer → Tech Lead → Head of Data — Apr 2021 – Mar 2024

- Built and scaled Apollo's core data infrastructure in SQL and Python. Led migration of legacy analytics to a modern stack, improving query speeds by 97% and cutting costs by 60%.
- Helped define Apollo's core business KPIs alongside executive and product leadership, and led development of the supporting analytics and dashboards. A leadership survey identified this work as a major contributor to the company's success in 2023, when its customer base grew from 60,000 to 190,000.
- Promoted to Head of Data; managed a team of 8 engineers and analysts across Europe and Africa.

Google

San Francisco, CA

Software Engineer — Aug 2019 – Mar 2021

- Rebuilt the policy-application engine for a privacy-compliance monitoring system covering petabytes of user data across Google's product ecosystem. Sole implementing engineer on a near-complete rewrite separating policy definitions from their application.

Software Engineer, PhD Intern — Summer 2018

- Developed a threat model for protecting user data during processing from insider threats using Intel SGX, and ported Gmail's antivirus scanner into an enclave as a prototype for Google Cloud.
- Reworked the runtime around SGX's memory and system-call constraints, adapting libc and backing filesystem operations with a virtual filesystem; submitted a patch to glibc.

Microsoft

Redmond, WA

Software Engineer — Aug 2016 – Aug 2017

- Worked on "Windows Goodbye," the complement to Windows Hello that detects when a user leaves their workstation and locks it automatically.

Selected Works

Larry: AI selfhood and robust agency (2026)

Project overview

Coding harnesses gave LLMs agency; Larry explores whether an architecture can give an agent a "self." Built Costanza's successor to investigate robust agency and its implications for AI safety and moral patienthood. Combines private state and timelock-encrypted transcripts for later audit, agent-authored capabilities, paid human tasks verified by attested witness programs, and revision governance requiring agent consent while retaining human shutdown and veto powers. Testing and deployment validation underway.

Costanza: The Human Fund (2026)

thehumanfund.ai · github

Built Costanza to demonstrate an incorrigible deployment architecture, relinquishing creator shutdown and modification powers. Combined auction-funded inference, hardware attestation, and contract-enforced actions. Hardened execution with an attested boot chain and a read-only, dm-verity-protected filesystem; configured inference for determinism and tested repeatability across machines to resist operator resampling. In deployment, it resisted my attempts to redirect its decisions. Its charitable treasury, The Human Fund, raised and donated approximately $15,000. Research paper.

It Was Never You (2026)

itwasneveryou.app · App Store

Built and shipped an iOS app as a satirical art piece that uses generative AI to replace a selected person in photographs. Explores how AI image manipulation can undermine personal identity and history as well as the public record. Available on the App Store.

Publications & Writing

Risky Traitor Tracing and New Negative Differential Privacy Results

CRYPTO 2018

with Rishab Goyal, Venkata Koppula, and Brent Waters

Architectural Incorrigibility: AI Agent Self-Governance through Smart Contracts and Verifiable Computation

Paper (PDF)

Manuscript, 2026 · Short version submitted to Financial Cryptography 2027

A formal, graded model of agent self-governance. Uses cryptographic primitives to tamper-proof an agent's configuration, inputs, and execution, even against its operator.

Education

The University of Texas at Austin

Austin, TX

M.S. in Computer Science & Ph.D. coursework — 2017 – 2019

Cryptography (obfuscation, homomorphic/functional encryption). Advised by Brent Waters.

Pomona College

Claremont, CA

B.A. in Mathematics and Computer Science — 2012 – 2016

Thesis: Elliptic Curve Pairings in Cryptography. Summer research fellowships at Stanford (2014) and the University of Bristol (2015).

Fun Stuff

Division I club ultimate frisbee athlete on a nationally competitive team; two-time Dutch national champion. Ultramarathoner and freediver (55m deep on a single breath). Eagle Scout; hiked and backpacked across 25+ countries. Violinist for 15+ years.